Math 10C — Unit 10: Assignments

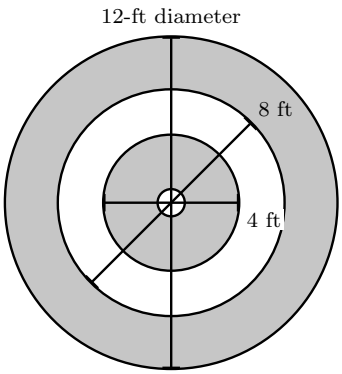
The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

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1. For each object below: state a referent that could be used to determine the *imperial* measure, use that referent to determine an approximate measurement, then determine the actual measurement using a ruler, metre stick, or tape measure.
(a) the length of a bookshelf (b) the height of a classmate (c) the width of a laptop screen

2. In curling, a house has three rings around a central button: the 12-foot circle, the 8-foot circle and the 4-foot circle, named for their *diameters* (drawn across each circle below), so their radii are 6 ft, 4 ft and 2 ft. Two players cannot tell by eye which of their two stones is closer to the button. Describe and explain a personal strategy that could be used to decide which stone is closer if



(a) the two stones are both in the outer ring, between the 12-foot and 8-foot circles.

(b) the two stones are both near the button, inside the 4-foot circle.

3. Match each approximate measurement on the left with the referent on the right.

i. ____ 1 inch	A. width of an adult pinky fingernail
ii. ____ 1 metre	B. thickness of a dime
iii. ____ 1 foot	C. width of five adult toes
iv. ____ 1 millimetre	D. diameter of a hockey puck
v. ____ 50 metres	E. width of a thumb across the joint
vi. ____ 20 centimetres	F. length of an adult foot
vii. ____ 2 metres	G. height of a standard doorway
viii. ____ 7.6 centimetres	H. width of a workbook
ix. ____ 0.1 metres	I. height of a 4-year-old child
x. ____ 1 centimetre	J. length of an adult pace
xi. ____ 1 yard	K. length of an Olympic swimming pool

4. Use a personal referent to estimate the length of your pencil case. Check the accuracy of your answer by measuring with a ruler.

5. Provide a personal referent for each of the following linear measurements, different from any referent already given in this lesson.

(a) millimetre

(b) centimetre

(c) metre

(d) inch

(e) foot

(f) yard

6. Describe and explain a personal strategy that could be used to determine the circumference of a basketball.

7. In the past, 10 city blocks was a common referent for a mile. What assumption is being made in stating that 10 city blocks is about 1 mile?

Questions 8–14 ***(Multiple Choice).*** Select the unit that would most reasonably be used to state each measurement.

8. Length of a hockey rink

A. millimetre B. mile
C. metre D. kilometre

9. Distance between two provincial capital cities

A. millimetre B. metre
C. foot D. kilometre

10. Diagonal measurement of a laptop screen

A. mile B. metre
C. inch D. millimetre

11. Length of a paperclip

A. millimetre B. mile
C. metre D. yard

12. Length of a school hallway

A. centimetre B. mile
C. yard D. inch

13. Width of a smartphone

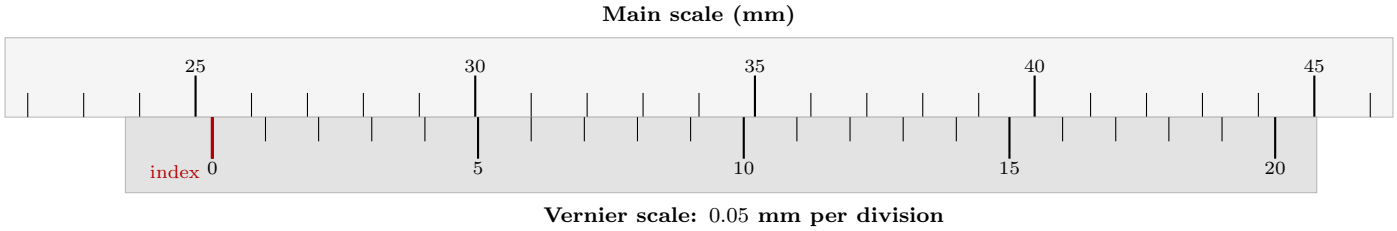
A. millimetre B. yard
C. metre D. mile

14. Height of a giraffe
A. millimetre B. inch C. mile D. metre

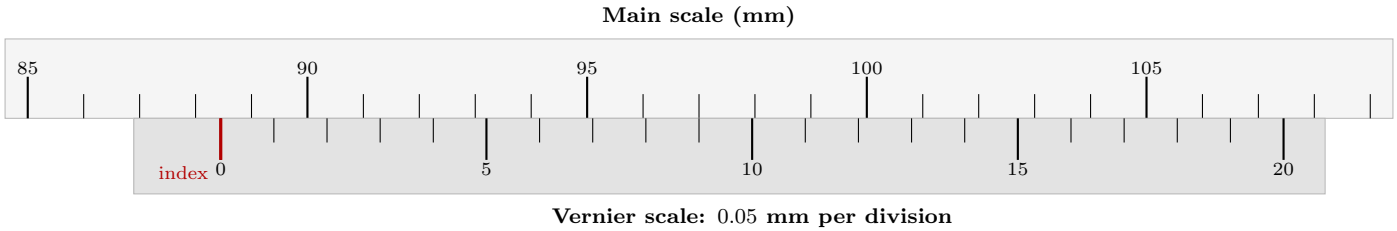
15. *(Numerical Response)* The length of an Olympic swimming pool is approximately 50 metres. The number of pool lengths that would have to be used as a referent to measure one kilometre is _____.

Record your answer left to right. | | | | |

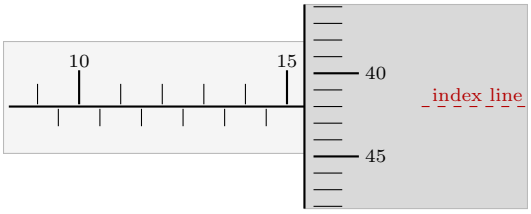
16. Determine each vernier caliper reading, in millimetres, to the caliper's precision.
(a)



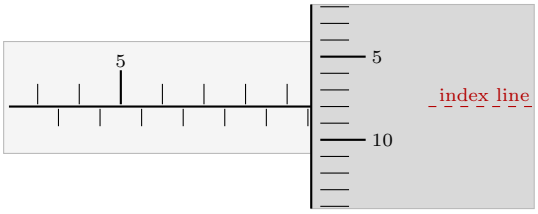
(b)



17. Determine each micrometer reading, in millimetres.
(a)



(b)



18. **Matching.** Match each object with the best measuring instrument. Each instrument may be used once, more than once, or not at all.

(i) The diameter of a pencil's lead _____ A. Trundle wheel
(ii) The length of a running track _____ B. Micrometer
(iii) The width of a textbook cover _____ C. Ruler
(iv) The depth of a small drilled hole _____ D. Vernier caliper

Lesson 2 Assignment *Show your work.*

1. Use the metric number line to convert each of the following to the indicated unit.
(a) 4250 cm = _____ m (b) 27 km = _____ cm (c) 63 mm = _____ cm
(d) 0.09 mm = _____ m (e) 0.72 m = _____ km (f) 15.84 cm = _____ mm

2. Convert each of the following to the indicated unit.
(a) 47 kL = _____ L (b) 763 mL = _____ L (c) 5384 g = _____ kg

3. On a map of British Columbia, 5 cm represents 84 km. If the actual distance between Kelowna and Kamloops is 126 km, use proportional reasoning to calculate the distance between the two cities on this map.

4. Convert each of the following using **unit analysis**. Show all work.
(a) 34 km = _____ cm (b) 56 700 mm = _____ km

5. Convert each rate to the indicated unit using **unit analysis**. Show all work.
(a) 180 m/s = _____ km/min (b) 6.2 m/s = _____ km/h

(c) 108 km/h = _____ m/s

6. *(Numerical Response)* Meera buys a spool of ribbon 9.6 m long for a craft project. The project requires the ribbon to be cut into 8 cm lengths. The number of 8 cm lengths she will have is _____.

Record your answer left to right. | | | | |

7. A hiking map has a scale of 1 : 50 000.
(a) Two campsites are 6.4 cm apart on the map. How far apart are they on the ground, in kilometres?
(b) A trail is 4.5 km long. How long is it on the map, in centimetres?

8. Convert the following.
(a) 4 ft 9 in to inches (b) 8 yd 2 ft to feet (c) 3 mi 210 yd to yards (d) 90 in to feet and inches

(e) 100 ft to yards and feet (f) 22 000 ft to miles and feet (g) 5 yd 1 ft 8 in to inches (h) 130 in to yards, feet, and inches

9. Convert, using proportional reasoning or unit analysis.
(a) 3.6 mi to in (b) 250 000 in to mi (to the nearest hundredth of a mile)

10. The Iron Trail ultra-marathon is run over a distance of 31 miles and 440 yards. Convert this distance to (a) yards (b) feet (c) inches

11. Using unit analysis, convert a speed of 45 miles per hour to feet per second (to the nearest tenth). Show the cancellation of units.

12. Convert the following.
(a) 12 ft 7 in = _____ m (two decimal places)
(b) 5.60 m = _____ ft _____ in (nearest inch)

13. **Estimate first, then convert.** For (a)–(d), use a benchmark to estimate mentally, then convert exactly (to the stated precision) and say whether your answer is reasonable.
(a) 6 ft 2 in to m (two decimal places) (b) 120 km to mi (one decimal place) (c) 8 in to cm (one decimal place)

(d) 50 m to yd (one decimal place) (e) Jordan converted 12 in and got 4.7 cm. Use a benchmark to explain why his answer cannot be right, then find the correct value.

14. Convert each height to the nearest hundredth of a metre, and each mass to the nearest kilogram.
(a) Connor McDavid's height, 6′1″ (b) Yao Ming's height, 7′6″ (c) Serena Williams's height, 5′9″

(d) a bull bison, about 2000 lb (e) a large beaver, about 44 lb (f) a 16-lb bowling ball

15. Convert each height to feet and inches to the nearest inch, and each mass to the nearest pound.
(a) Andre De Grasse's height, 1.76 m (b) Peter Crouch's height, 2.01 m (c) Usain Bolt's height, 1.95 m

(d) a dairy cow, about 680 kg (e) a husky, about 25 kg (f) a full hockey bag, about 18 kg

16. Some track meets historically ran the half-mile (880 yards) event, now replaced by the 800 metre race. Which distance, half a mile or 800 metres, is longer, and by how much? Answer in metres and in yards, to the nearest whole number.

17. The 220-yard dash, a historical sprint distance, has been replaced by the 200 metre race. Which distance, 220 yards or 200 metres, is longer, and by how much? Answer in metric units and in imperial units, to the nearest whole number.

18. *(Multiple Choice)* The number of inches in 4 yards and 9 inches is
A. 57 **B.** 144 **C.** 153 **D.** 21

19. *(Numerical Response)* A car crosses the Canada–US border. Its speedometer reads 95 km/h. To the nearest whole number, this speed in mph is _____.
Record your answer left to right. ☐☐☐☐

20. *(Numerical Response)* A vehicle is travelling at 62 mph. To the nearest whole number, this speed in km/h is _____.
Record your answer left to right. ☐☐☐☐

Extension — the speed of light (Questions 21–23)

Use the following information to answer Questions 21–23.

In physics, the speed of light in a vacuum is denoted by the letter c. Its value is recorded as exactly 299 792 458 m/s.

21. *(Extension)* c is often approximated in kilometres per second. State the speed of light to the nearest one hundred thousand kilometres per second.

22. *(Extension; Multiple Choice)* In the imperial system, c is usually estimated in miles per second. Determine the speed of light to the nearest thousand miles per second.
A. 186 000 **B.** 168 000
C. 1 860 000 **D.** 300 000

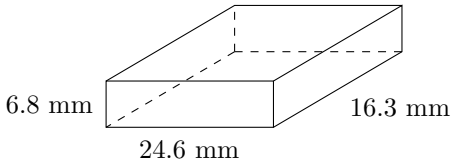
23. *(Extension; Multiple Choice)* It takes light approximately 150 000 years to cross the Andromeda galaxy. Assuming an average of 365.25 days per year, the approximate distance in miles across Andromeda is
A. 8.8×10^{12} **B.** 8.8×10^{17}
C. 1.4×10^{13} **D.** 1.4×10^{18}

Lesson 3 Assignment *Show your work.*

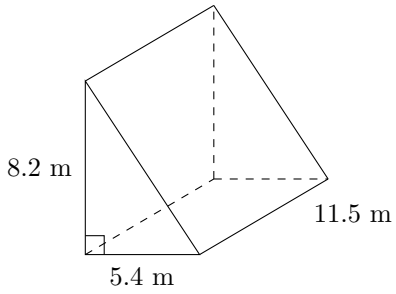
1. Convert the following volumes to the indicated unit. Round to the nearest thousandth where necessary.
(a) 2148 cm^3 to litres (b) 3.8 litres to cm^3 (c) 512.75 m^3 to litres

(d) 35 000 litres to m^3 (e) 60 US gallons to litres (f) 750 litres to UK gallons

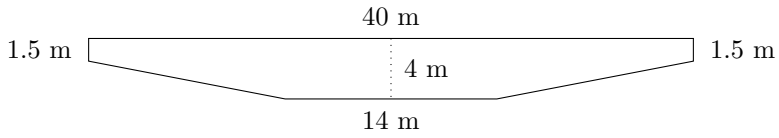
2. Calculate the surface area and volume of each right prism. Round to the nearest tenth where necessary.
(a)



(b)



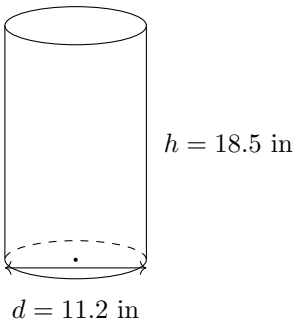
3. A cross section of the water in a swimming pool is shown below.



(a) Calculate the area of the cross section.

(b) If the pool is 22 metres wide, determine the number of litres of water it holds when full.

4. Calculate the volume and surface area of the cylinder shown. Answer in appropriate units to the nearest tenth.



5. Chemical waste is stored in metal drums with a diameter of 52 cm and a height of 90 cm. Sketch a drum in the box and label its dimensions before you start.

Sketch

(a) Determine the volume of the drum to the nearest tenth of a litre.

(b) Determine the volume of the drum in US gallons, to the nearest tenth.

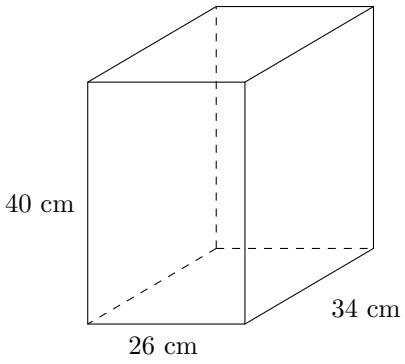
(c) The exterior of the drum, except the base, is to be painted blue. Determine how many litres of paint are needed to paint 500 drums if one litre of paint covers 9.5 m^2. Answer to the nearest tenth of a litre.

6. A cube has a volume of 0.729 litres. Determine the length of each side of the cube in cm.

7. A cylindrical jar of jam has a volume of 452 cm^3. If the height of the jar is 6.4 cm, determine the area of the bottom of the jar to the nearest tenth of a cm^2.

8. A cylindrical soda can has a volume of 355 mL. Determine, to the nearest 0.1 cm, the diameter of the can if the height is 11.5 cm.

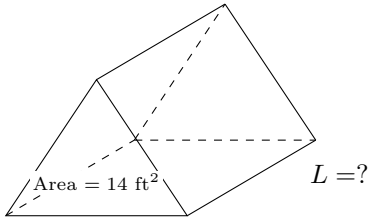
9. A wooden display case is built in the shape of a rectangular prism with outer dimensions 34 cm (depth) × 26 cm (width) × 40 cm (height). The wood used is 2 cm thick on all sides.



(a) Determine the volume of the interior space of the case.

(b) All outer faces except the front face (26 cm × 40 cm) are to be varnished. Determine the surface area to be varnished.

10. A pup tent in the shape of a triangular prism has a triangular entrance with an area of 14 ft^2. If the space inside the tent has a volume of 2.8 m^3, determine the length of the tent to the nearest foot.

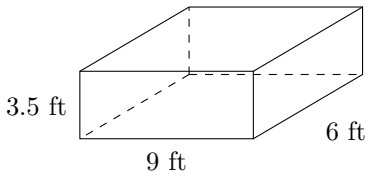


11. *(Multiple Choice)* Yin cuts the lid off a cylindrical can. Which expression gives the surface area of the remaining can (curved side and bottom only, no top)?

A. $2\pi r^2 + 2\pi rh$ **B.** $\pi r^2 + 2\pi rh$ **C.** $\pi r^2 + \pi rh$ **D.** $2\pi r(r + h)$

Use the following information for Questions 12–14.

A delivery truck carries gravel in an open-top box shaped like a right rectangular prism, 9 ft long, 6 ft wide, and 3.5 ft high. The box has no top; a tarp covers the gravel once loaded.



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12. *(Multiple Choice)* The volume of gravel the truck can transport, in m^3, is closest to

A. 0.03 m^3 **B.** 5.4 m^3 **C.** 17.6 m^3 **D.** 57.6 m^3

13. *(Numerical Response)* Before the box is attached to the truck, all of the inside *and* outside faces of the box are coated with primer. The amount of primer needed, to the nearest 0.1 m^2, is _____.

Record your answer left to right. ☐☐☐☐

14. *(Numerical Response)* Designers want a new box that holds 12 m^3 of gravel, keeping the same 9 ft length and 6 ft width — only the height changes. The new height can be written as a feet and b inches, to the nearest inch. The value of $a + b$ is _____.

Record your answer left to right. ☐☐☐☐

15. *(Numerical Response)* To the nearest mL, the number of mL in a US pint is _____.

Record your answer left to right. ☐☐☐☐

16. Work backwards from the surface area to find the unknown dimension.

(a) A closed can has a surface area of 400 cm^2 and a radius of 4 cm. Determine its height, to the nearest tenth of a centimetre.

(b) A closed box has a square base 10 cm on a side and a surface area of 520 cm^2. Determine its height.

Lesson 4 Assignment *Show your work.*

Unless stated otherwise, "pyramid" means right pyramid and "cone" means right circular cone. Use the π key, carry full precision, and round once at the end.

1. Each part describes one solid that "just fits" inside another (identical base, identical height). Use the $\frac{1}{3}$ volume relationship to state the requested volume — no formula needed.

(a) A cylinder has a volume of 36 ft^3. State the volume of a cone that just fits inside the cylinder.

(b) A cone has a volume of 9 m^3. State the volume of a cylinder that just holds the cone.

(c) A rectangular prism has a volume of 240 in^3. State the volume of a rectangular pyramid that just fits inside the prism.

(d) A pentagonal right pyramid has a volume of 45 mm^3. State the volume of a pentagonal right prism that just holds the pyramid.

2. In each part, make a sketch of the figure and calculate its volume to the nearest whole number.

(a) a cone with base radius 9 mm and height 12 mm

(b) a square-based pyramid with side 24 cm and height 16 cm

(c) a cone with base diameter 20 ft and slant height 26 ft

(d) a cone with slant height 37 in and vertical height 35 in

(e) a triangular-based pyramid with base area 42 m^2 and height 9 m

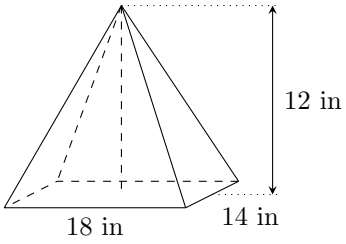
3. Calculate the surface area of the figures in Question 2, parts (a)–(c). Answer to the nearest whole number.

(a)

(b)

(c)

4. Consider the rectangular-based pyramid shown, with a height of 12 in.



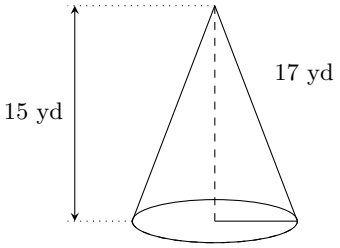
(a) Sketch the net of the pyramid, marking the base dimensions (18 in by 14 in) and both slant heights on your net. (The 12 in height is inside the solid, so it is not an edge of the net.)

(b) Determine the surface area of the pyramid to the nearest square inch.

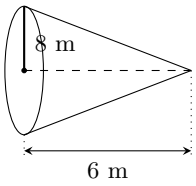
(c) Determine the volume of the pyramid.

5. Determine the volume and surface area of each cone below. Answer to one decimal place.

(a)



(b)



6. Cups Incorporated makes conical paper cups for water dispensers. One of its cups has a diameter of 9 cm, a height of 11 cm, and no base.

(a) Determine the slant height of the cup to the nearest 0.1 cm.

(b) To the nearest whole number, how many mL of water will the cup hold?

(c) Determine how much paper is required, to the nearest square cm, to make the cup.

7. Devon constructs a cube with interior edges of 12 cm.

(a) Determine the volume of the largest possible pyramid that will just fit inside the cube.

(b) Determine, to the nearest cubic centimetre, the volume of the largest possible cylinder that will just fit inside the cube.

(c) Determine, to the nearest cubic centimetre, the volume of the largest possible cone that will just fit inside the cube.

8. An engineering firm designs a ceremonial mound in the shape of a square-based pyramid, with a base 300 m by 300 m and a height of 50 m. Determine the volume of the pyramid in cubic metres.

9. A pyramid-shaped grain storage monument has a volume of 972 000 m^3 and a height of 90 m. The base is square. Determine the length of each side of the base to the nearest metre.

10. Determine the exact volume and surface area of a right pyramid with a height of $\sqrt{5}$ feet and a hexagonal base with six sides of 4 feet.

11. An architecture firm designs two square-based glass pyramid atriums. Atrium A has a height of 60 ft and a square base with sides of 90 ft; it is covered with 480 rhombus-shaped and 60 triangular glass panels (the base is not glass). Atrium B has a height of 45 ft and a square base with sides of 60 ft.

(a) To the nearest hundred square feet, how many square feet of glass are in Atrium A?

(b) How many times as great as the volume of Atrium B is the volume of Atrium A? Answer to the nearest tenth.

12. Sand dropped from a conveyor belt forms a conical pile with a diameter of 18.0 ft and a height of 7.5 ft. Determine the volume of the sand pile to the nearest 0.01 m^3.

13. *(Multiple Choice)* A square-based pyramid has each side length tripled, but the height remains the same. The volume of the original pyramid is what fraction of the volume of the new pyramid?

A. $\frac{1}{3}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{9}$ **D.** $\frac{1}{27}$

14. *(Multiple Choice)* A cone has its radius tripled and its height reduced to $\frac{1}{3}$ of its original value. Which statement is correct?

A. The new volume equals the original volume.
B. The new volume is twice the original volume.
C. The new volume is three times the original volume.
D. None of the above is correct.

15. *(Numerical Response)* An engineer is designing a cone-shaped storage silo with a base radius of 12 m that is to hold 5000 m^3 of grain. The height of the silo, to the nearest tenth of a metre, is _____.

Record your answer left to right. | | | | |

16. *(Numerical Response)* A rectangular-based pyramid has a base length of 6 ft and a height of 15 ft. It has a volume of 270 ft^3. The base width of the pyramid, to the nearest centimetre, is _____.

Record your answer left to right. | | | | |

17. A closed cone has a surface area of 300 cm^2 and a base radius of 6 cm.

(a) Determine the slant height s, to the nearest tenth of a centimetre.

(b) Use s to determine the perpendicular height h, to the nearest tenth of a centimetre.

18. A grain silo is a cylinder with radius 3 m and height 10 m, topped by a cone with the same radius and a height of 2 m. The silo stands on the ground. Sketch the silo and label its dimensions.

(a) Determine the volume of the silo, to the nearest tenth of a cubic metre.

(b) Determine the area of the outside of the silo that must be painted (walls and roof, not the ground), to the nearest tenth of a square metre.

(c) Explain why the circle where the cone meets the cylinder is not included in (b).

Lesson 5 Assignment *Show your work.*

1. Calculate the following to the nearest tenth.

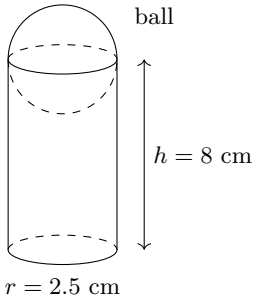
(a) the surface area of a basketball with a diameter of 24 cm

(b) the volume of a decorative globe with a radius of 10 in

(c) the volume and surface area of a (fictional) moon named Xylo, if Xylo is spherical with a diameter of 10 000 km (you may write the large answers in scientific notation)

(d) the volume and surface area of a hemisphere with a radius of 3.6 m

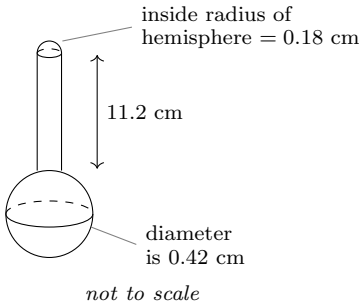
2. The diagram shows a roll-on deodorant: a cylindrical container of radius 2.5 cm and height 8 cm, into which a ball of the same radius is set. The centre of the ball is level with the top of the container, so half of the ball is inside the container. Liquid deodorant completely fills the space below the ball. Calculate, to the nearest mL, the volume of liquid deodorant in the container.



3. (a) Solve the equation $V = \frac{4}{3}\pi r^3$ for r. Write the answer both as a radical and as a power.

(b) The volume of a beach ball is 65 000 cm^3. Calculate its radius to the nearest tenth of a centimetre.

4. A model mercury thermometer consists of a spherical bulb, topped by a thin cylindrical stem, topped by a small hemispherical cap. The stem has the same inside radius as the cap, 0.18 cm. Use the dimensions shown (not to scale) to answer the questions below, to the nearest hundredth.



not to scale

(a) Determine the volume of the space inside the thermometer.

(b) The thermometer is made of glass. Determine the surface area of glass required to build it. Assume the missing glass where the stem meets the bulb equals the area of the base of the cylindrical stem.

5. Priya designs a pencil case in the shape of a hemisphere and plans to sew 400 of them. The fabric costs five cents per hundred square centimetres, and each zipper costs six cents. Each case uses exactly one zipper, and adding the zipper does not change the amount of fabric needed. The diameter of each case is 18 cm.

(a) Calculate the surface area of each case, to one decimal place.

(b) Determine the total cost of making all 400 pencil cases.

6. The surface area of a spherical fuel tank is 380 m^2. Determine its volume to the nearest cubic metre.

7. A bowling ball has a surface area of 254.5 in^2. Determine its volume to the nearest tenth of a cubic inch.

8. A silver miniature soccer ball is packaged as an award inside a cubic container that just fits the ball. The container has a volume of 2197 cm^3.

(a) Calculate the surface area of the silver ball, to the nearest cm^2.

(b) Once the ball is inside the cube, bubble wrap fills 75% of the remaining space in the container. Determine the volume of bubble wrap used, to the nearest cm^3.

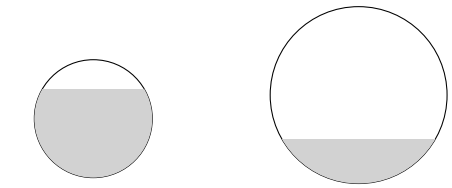
9. *(Multiple Choice)* Chocolate Rules Ice Cream Company scoops a spherical ball of ice cream with a volume of 90 mL onto a cone. Which cone diameter would best fit the scoop?
A. 3 cm **B.** 4.5 cm **C.** 5.5 cm **D.** 7 cm

10. *(Multiple Choice)* The surface area of a sphere is 415 m^2. Its radius, to the nearest tenth of a metre, is
A. 33.0 **B.** 11.5 **C.** 5.7 **D.** 2.9

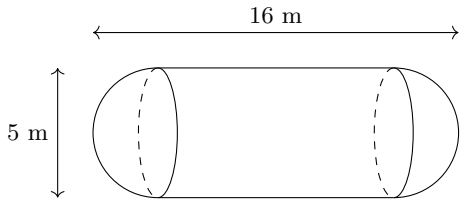
11. *(Numerical Response)* A sphere has a radius of 21 mm. The volume of the sphere, in cm^3 to the nearest tenth, is _____.
Record your answer left to right. | | | | |

12. *(Numerical Response)* A cylindrical jar has diameter 8 cm and contains water to a depth of 6 cm. A sphere of diameter 4 cm is dropped into the jar and sinks to the bottom. The new water depth, to the nearest tenth of a centimetre, is _____.
Record your answer left to right. | | | | |

13. *(Numerical Response)* Sphere A is 75% full of water and Sphere B is 25% full of water. If both spheres contain an equal volume of water, the diameter of Sphere B, to the nearest tenth of a centimetre, is _____.
Record your answer left to right. | | | | |



14. *(Numerical Response)* A boiler is shaped like a cylinder with a hemispherical end cap on each side. Its total length is 16 m and its diameter is 5 m. Its cubic content, to the nearest m^3, is _____.
Record your answer left to right. | | | | |



Use the following information to answer questions 15 and 16.

A solid gold earring is shaped like a hemisphere topped by a cone.
The diameter of the hemisphere and the height of the cone are each 7.4 mm.
The mass of 1 cubic centimetre of gold is 19.3 grams.

15. *(Numerical Response)* The mass of the earring, to the nearest tenth of a gram, is _____.
Record your answer left to right. | | | | |

16. *(Numerical Response)* The surface area of the earring, to the nearest square millimetre, is _____.
Record your answer left to right. | | | | |

Lesson 6 Assignment *Show your work.*

1. *(Multiple Choice)* Which is the most suitable imperial unit of measurement for the length of a pencil?
A. mile **B.** yard **C.** foot **D.** inch

2. *(Multiple Choice)* A rectangular photograph measures 62 cm in width and 1.24 m in length. The difference between these two measurements, in mm, is
A. 6.2 **B.** 62 **C.** 620 **D.** 6200

3. *(Multiple Choice)* The number of kilometres in 0.036 centimetres is
A. 0.0000000036 **B.** 360 **C.** 0.00000036 **D.** 36

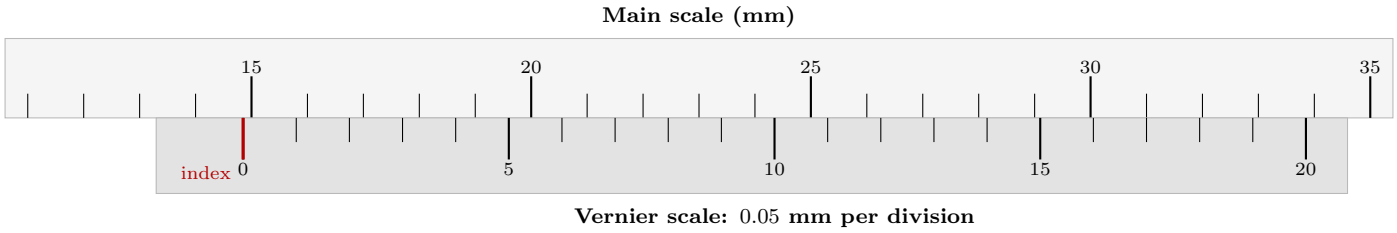
4. *(Multiple Choice)* Consider the following four statements.
Statement 1: 1 mm = 0.1 cm Statement 2: 1 m = 100 cm Statement 3: 1 km = 100 000 cm
Statement 4: 1 cm = 0.01 mm
How many of these statements are false?
A. zero **B.** one **C.** three **D.** four

5. *(Multiple Choice)* The height of horses is often measured in hands. Using the width of an adult hand as a referent for 4 inches, the height of a horse measuring $15\frac{3}{4}$ hands is
A. $4\frac{1}{2}$ feet **B.** 5 feet **C.** $5\frac{1}{4}$ feet **D.** $6\frac{1}{4}$ feet

6. *(Multiple Choice)* Which of the following measuring instruments would be easiest to use to measure the width of a small keyhole?
A. micrometer **B.** vernier caliper **C.** ruler **D.** trundle wheel

7. *(Multiple Choice)* A section of the Trans-Canada Highway is 968 km long. Using the approximate conversion 5 miles = 8 km, this distance in miles is
A. 242 **B.** 605 **C.** 788 **D.** 1548.8

8. *(Multiple Choice)* The vernier caliper below reads to the nearest 0.05 mm. The reading is



A. 14.85 mm **B.** 14.17 mm **C.** 15.85 mm **D.** 14.80 mm

9. *(Numerical Response)* Priya and Devon are tiling a rectangular hallway floor that is 6 yards long and 4 yards wide, using square tiles that measure 9 inches by 9 inches. The number of tiles they need is _____.

Record your answer left to right. ☐☐☐☐

10. *(Multiple Choice)* The scale on a map is 1 : 850 000. If the actual distance between two towns is 425 km, the distance between them on the map is

A. 5 cm **B.** 50 cm
C. 500 cm **D.** 0.5 cm

11. *(Numerical Response)* An aquarium has dimensions 60 cm by 45 cm by 20 cm. The directions on a bottle of fish medicine are to add 5 drops per 10 litres of water. The number of drops that should be added to a completely full aquarium is _____.

Record your answer left to right. ☐☐☐☐

12. *(Numerical Response)* Mia is marking the sideline of a field, which needs to be 10 yards long. Her shoe is 9 inches long, so each heel-to-toe step covers 9 inches; she uses this as a referent, placing her feet heel to toe repeatedly along the line. The number of times she must place her feet is _____.

Record your answer left to right. ☐☐☐☐

13. *(Multiple Choice)* A hex key set includes metric sizes 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, and 8 mm, and imperial sizes $\frac{1}{8}$ in, $\frac{5}{32}$ in, $\frac{3}{16}$ in, $\frac{7}{32}$ in, $\frac{1}{4}$ in, and $\frac{5}{16}$ in. Dov's 5 mm key is too small for a bolt and his 6 mm key is too large. Which imperial key should he try?

A. $\frac{3}{16}$ in **B.** $\frac{7}{32}$ in **C.** $\frac{1}{4}$ in **D.** $\frac{5}{16}$ in

14. *(Multiple Choice)* Using the same hex key set as Question 13, Dov's $\frac{1}{4}$ in key is too small for a different bolt and his $\frac{5}{16}$ in key is too large. Which metric key should he try?

A. 5 mm **B.** 6 mm **C.** 7 mm **D.** 8 mm

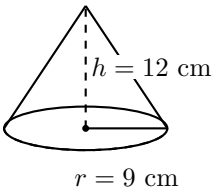
15. *(Numerical Response)* A sprinter completes a 200 m time trial in 11.82 seconds. Her average speed, to the nearest 0.1 km/h, is _____.

Record your answer left to right. ☐☐☐☐

16. *(Numerical Response)* A cylindrical can of paint has a base diameter of 8.2 cm and a height of 12.5 cm. A paper label completely covers the curved surface of the can. The surface area of the label, to the nearest cm^2, is _____.

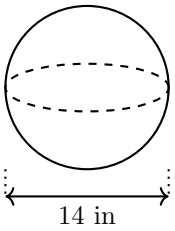
Record your answer left to right. ☐☐☐☐

17. *(Multiple Choice)* A conical funnel has a base radius of 9 cm, a height of 12 cm, and no base (it is open at its wide end). The surface area of the funnel, in terms of π, is



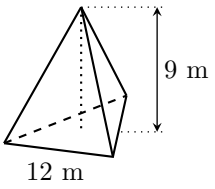
A. 81π
B. 135π
C. 216π
D. 225π

18. *(Multiple Choice)* A spherical ornament has a diameter of 14 inches. The surface area of the ornament, to the nearest square inch, is



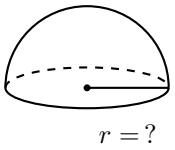
A. 308
B. 616
C. 1232
D. none of the above

19. *(Multiple Choice)* A decorative object is in the shape of a triangular pyramid. The base is an equilateral triangle with sides of 12 m, and the height is 9 m. The volume of the pyramid in m^3 is



A. $108\sqrt{3}$
B. $36\sqrt{3}$
C. 324
D. 216

20. *(Multiple Choice)* The volume of a hemispherical water tank is 718.4 m^3. The radius of the tank, to the nearest tenth of a metre, is



A. 5.0
B. 7.0
C. 9.0
D. 12.0

21. *(Written Response)* Golden Plains Dairy stores fresh milk in a cylindrical vat with a radius of 2.2 m and a height of 4.5 m, filled completely.

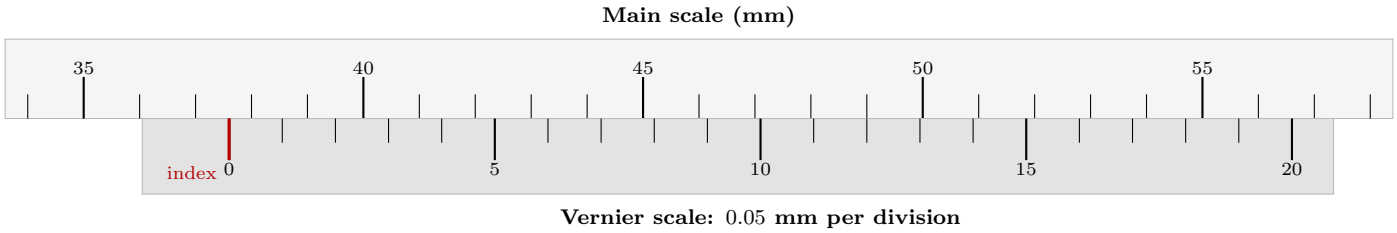
(a) Determine the volume of milk in the vat, correct to one decimal place.

(b) The milk is transported to a processing plant using rectangular tanker trucks with interior dimensions 2.5 m by 2 m by 8 m. Using the volume from (a), determine the number of trucks needed.

(c) At the plant, the milk is poured into a hemispherical storage tank with a diameter of 10 m. Using the total volume from (a), determine what percentage of the hemispherical tank's capacity is filled, to the nearest tenth of a percent.

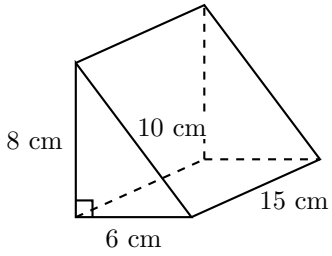
(d) Two trucking companies bid for the job. Foothills Transport charges C$200 per truck plus C$3.50 per m^3 of milk. Rocky Mountain Haulers charges US$150 per truck plus US$0.10 per ft^3 of milk. Using C$1 = US$0.93 and 2.54 cm = 1 inch, and using the volume and number of trucks from (a) and (b), determine which company is cheaper.

22. *(Written Response)* State the reading of this vernier caliper, accurate to 0.05 mm.



23. *(Written Response)* A chocolate bar is a right triangular prism. Its triangular ends are right triangles with legs of 6 cm and 8 cm and a hypotenuse of 10 cm, and the bar is 15 cm long.

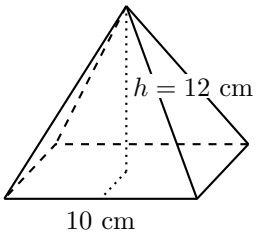
(a) Determine the volume of the bar.



(b) Determine the surface area of the foil wrapper that covers the bar exactly.

(c) Melted, the chocolate is poured into a measuring cup. State its volume in millilitres and in litres.

24. *(Written Response)* A monument is a square-based pyramid. Each side of the base is 10 cm and the pyramid's perpendicular height is 12 cm.

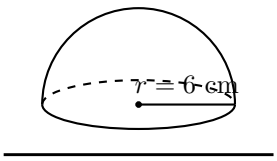


(a) Determine the slant height s of a triangular face.

(b) Determine the total surface area of the monument, including its base.

(c) Determine the volume of the monument.

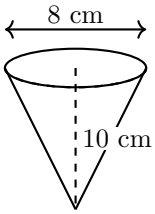
25. *(Multiple Choice)* A solid glass paperweight is a hemisphere of radius 6 cm, resting flat side down. Its total surface area, including the flat circular base, to the nearest tenth of a cm^2, is



A. 226.2 cm^2 **B.** 339.3 cm^2
C. 452.4 cm^2 **D.** 904.8 cm^2

26. *(Numerical Response)* A cone-shaped paper cup has a diameter of 8 cm and a depth of 10 cm. Its capacity, to the nearest cm^3, is _____.

Record your answer left to right. | | | | |



Questions 27–29 — micro- and mega-prefixes (Extension), the micrometer

27. *(Extension; Multiple Choice)* Consider the following four statements.

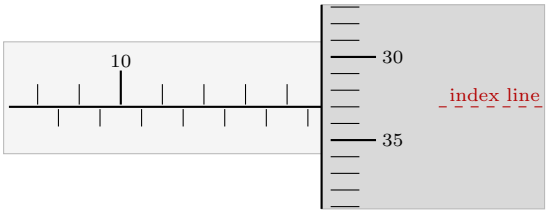
Statement 1: 1 mm = 0.1 cm Statement 2: 1 m = 100 cm Statement 3: 1 μm = 1000 nm

Statement 4: 1 Mm = 0.001 km

How many of these statements are false?

A. zero **B.** one **C.** three **D.** four

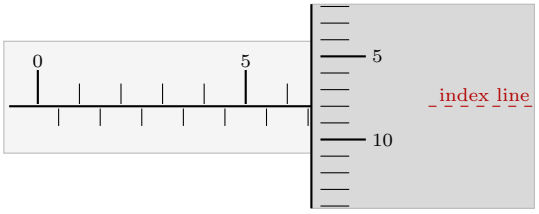
28. *(Multiple Choice)* The micrometer below has a measuring range of 0–25 mm and is accurate to 0.01 mm. The sleeve scale has increments of 0.5 mm and the thimble scale has increments of 0.01 mm. The reading is



Sleeve: a mark every 0.5 **mm**
Thimble: 0.01 **mm per division**

A. 14.83 mm **B.** 13.83 mm **C.** 14.33 mm **D.** 13.33 mm

29. *(Written Response)* State the reading of this micrometer, accurate to 0.01 mm.



Sleeve: a mark every 0.5 **mm**
Thimble: 0.01 **mm per division**

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. Answers vary; the estimate should land within roughly 10% of the measured value. Samples: (a) adult pace (≈ 1 yd) (b) adult pace or cubit laid off vertically (c) thumb width (≈ 1 in). **2.** (a) Walk normal paces (≈ 1 yd each) from each stone to the button; fewer paces = closer. (b) A pace (1 yd = 3 ft) is too coarse inside the 4-foot circle, so switch to hand-widths (4 in) or thumb-widths (1 in). **3.** i. E ii. I iii. F iv. B v. K vi. H vii. G viii. D ix. C x. A xi. J. **4.** Answers vary — a pencil case is typically 20–25 cm, so about one workbook width (≈ 20 cm) or about 8–10 thumb widths; verify with a ruler. **5.** Answers vary. Samples: (a) width of a strand of uncooked spaghetti (b) diameter of an AA battery (c) height of a kitchen countertop (d) width of a matchbox (e) length of a standard school ruler (12 in) (f) chin to outstretched fingertips. **6.** Wrap string around the widest part of the ball, mark it, straighten it, then count pinky-fingernail widths (≈ 1 cm each) along the string. **7.** That all 10 blocks are about the same length as one another, so each block is about $\frac{1}{10}$ mi (≈ 176 yd ≈ 160 m). **8.** C. **9.** D. **10.** C. **11.** A. **12.** C. **13.** A. **14.** D. **15.** 20 (pool lengths). **16.** (a) 25.30 mm (b) 88.45 mm. **17.** (a) 15.42 mm (b) 9.58 mm. **18.** (i) B (ii) A (iii) C (iv) D.

Lesson 2 Assignment

1. (a) 42.5 m (b) 2 700 000 cm (c) 6.3 cm (d) 0.000 09 m (e) 0.000 72 km (f) 158.4 mm. **2.** (a) 47 000 L (b) 0.763 L (c) 5.384 kg. **3.** 7.5 cm. **4.** (a) 3 400 000 cm (b) 0.0567 km. **5.** (a) 10.8 km/min (b) 22.32 km/h (c) 30 m/s. **6.** 120. **7.** (a) 3.2 km (b) 9 cm. **8.** (a) 57 in (b) 26 ft (c) 5490 yd (d) 7 ft 6 in (e) 33 yd 1 ft (f) 4 mi 880 ft (g) 200 in (h) 3 yd 1 ft 10 in. **9.** (a) 228 096 in (b) 3.95 mi. **10.** (a) 55 000 yd (b) 165 000 ft (c) 1 980 000 in. **11.** 66.0 ft/s. **12.** (a) 3.84 m (b) 18 ft 4 in. **13.** (a) est. 1.85 m; 1.88 m (b) est. 75 mi; 74.6 mi (c) est. 20 cm; 20.3 cm (d) est. 55 yd; 54.7 yd (e) 12 in $\approx$ 30 cm, so 4.7 cm is far too small (he divided); 30.48 cm. **14.** (a) 1.85 m (b) 2.29 m (c) 1.75 m (d) 909 kg (e) 20 kg (f) 7 kg. **15.** (a) $5'9''$ (b) $6'7''$ (c) $6'5''$ (d) 1496 lb (e) 55 lb (f) 40 lb. **16.** The half-mile is longer, by about 5 m (about 5 yd). **17.** The 220-yard dash is longer, by about 1 m (about 1 yd). **18.** C. **19.** 59. **20.** 100. **21.** (Extension) 300 000 km/s. **22.** (Extension) A. **23.** (Extension) B.

Lesson 3 Assignment

1. (a) 2.148 L (b) 3800 cm^3 (c) 512 750 L (d) 35 m^3 (e) 227.124 L (f) 164.977 UK gal. **2.** (a) $SA = 1358.2$ mm^2, $V = 2726.7$ mm^3 (b) $SA = 313.6$ m^2, $V = 254.6$ m^3. **3.** (a) 127.5 m^2 (b) 2 805 000 L. **4.** $V = 1822.6$ in^3, $SA = 848.0$ in^2. **5.** (a) 191.1 L (b) 50.5 US gal (c) 88.6 L. **6.** 9 cm. **7.** 70.6 cm^2. **8.** 6.3 cm. **9.** (a) 23 760.0 cm^3 (b) 5528 cm^2. **10.** 7 ft. **11.** B. **12.** B. **13.** 29.5 m^2. **14.** 17. **15.** 473 mL. **16.** (a) 11.9 cm (b) 8 cm.

Lesson 4 Assignment

1. (a) 12 ft^3 (b) 27 m^3 (c) 80 in^3 (d) 135 mm^3. **2.** (a) 1018 mm^3 (b) 3072 cm^3 (c) 2513 ft^3 (d) 5278 in^3 (e) 126 m^3. **3.** (a) 679 mm^2 (b) 1536 cm^2 (c) 1131 ft^2. **4.** (a) net with $s_{18} = \sqrt{193} \approx 13.9$ in and $s_{14} = 15$ in (b) 712 in^2 (c) 1008 in^3. **5.** (a) $V = 1005.3$ yd^3, $SA = 628.3$ yd^2 (b) $V = 402.1$ m^3, $SA = 452.4$ m^2. **6.** (a) 11.9 cm (b) 233 mL (c) 168 cm^2. **7.** (a) 576 cm^3 (b) 1357 cm^3 (c) 452 cm^3. **8.** 1 500 000 m^3. **9.** 180 m. **10.** $V = 8\sqrt{15}$ ft^3, $SA = 24\sqrt{3} + 12\sqrt{17}$ ft^2. **11.** (a) 13 500 ft^2 (b) 3.0 times. **12.** 18.01 m^3. **13.** C. **14.** C. **15.** 33.2. **16.** 274. **17.** (a) 9.9 cm (b) 7.9 cm. **18.** (a) 301.6 m^3 (b) 222.5 m^2 (c) The circle where the cone sits on the cylinder is hidden inside the silo, so it is not part of the outside surface.

Lesson 5 Assignment

1. (a) 1809.6 cm^2 (b) 4188.8 in^3 (c) $V \approx 5.2 \times 10^{11}$ km^3, $SA \approx 3.1 \times 10^8$ km^2 (d) $V \approx 97.7$ m^3, $SA \approx 122.1$ m^2. **2.** 124 mL. **3.** (a) $r = \sqrt[3]{\dfrac{3V}{4\pi}} = \left(\dfrac{3V}{4\pi}\right)^{1/3}$ (b) 24.9 cm. **4.** (a) 1.19 cm^3 (b) 13.32 cm^2. **5.** (a) 763.4 cm^2 (b) \$176.68. **6.** 697 m^3. **7.** 381.8 in^3. **8.** (a) 531 cm^2 (b) 785 cm^3. **9.** C. **10.** C. **11.** 38.8 cm^3. **12.** 6.7 cm. **13.** 14.4 cm. **14.** 281 m^3. **15.** 4.1 g. **16.** 182 mm^2.

Lesson 6 Assignment

1. D. **2.** C. **3.** C. **4.** B. **5.** C. **6.** B. **7.** B. **8.** A. **9.** 384. **10.** B. **11.** 27. **12.** 40. **13.** B. **14.** C. **15.** 60.9 km/h. **16.** 322 cm^2. **17.** B. **18.** B. **19.** A. **20.** B. **21.** (a) 68.4 m^3 (b) 2 trucks (c) 26.1% (d) Rocky Mountain Haulers, C\$582.41 vs. C\$639.48. **22.** 37.60 mm. **23.** (a) 360 cm^3 (b) 408 cm^2 (c) 360 mL = 0.36 L. **24.** (a) $s = 13$ cm (b) 360 cm^2 (c) 400 cm^3. **25.** B. **26.** 168 cm^3. **27.** B (Extension). **28.** A. **29.** 6.58 mm.